Echoes and Delays: Time-to-Build in Production Networks

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Motivation

- Firms in a modern economy rely on a complex network of suppliers, whose time-to-build varies significantly
  - Ex.: $<$ 1 month for furniture assembly, 1-2 years for container ships

- Most of the literature on production networks ignores time-to-build and delays
  - Acemoglu et al. (2012), Baqaee and Farhi (2019, 2020), ... essentially static
  - Roundabout production: inputs are produced and used in the same period, so disruptions are resolved within the period

- How does the introduction of time-to-build or delivery lags affect the dynamics of production networks?

What We Do

- We propose a simple model to introduce time-to-build (T2B) in production networks
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- We analyze how T2B contributes to the propagation of shocks:
  1. Persistence, delays and bottlenecks $\Rightarrow$ shocks take much longer to work through the economy
  2. Echoes and endogenous fluctuations
     $\Rightarrow$ cycles in the network make shocks come back, creating cyclical fluctuations
  3. Dynamic sectoral comovements and aggregation
     $\Rightarrow$ cyclical fluctuations survive aggregation

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- We test the model's prediction in monthly U.S. industrial production data
  - customers move with their suppliers at the measured delays
  - aggregate implications visible in the data

Related Literature

- Shock propagation in production networks
  - Acemoglu et al. (2012), Barrot and Sauvagnat (2016), Acemoglu et al. (2017), Baqaee and Farhi (2019), Carvalho et al. (2021), Ghassibe (2024), ...
- Time-to-build
  - Kydland and Prescott (1982), Schwartzman (2014), ...
  - In production networks: Long and Plosser (1983), Carvalho and Reischer (2021), Liu and Tsyvinski (2024), Bizzarri (2024), Bizzarri, Pangallo and Queirós (2025), Leng et al. (2025)
- Endogenous fluctuations in multi-sector economies
  - Benhabib and Nishimura (1979, 1985, ...)
- Delays in supply chains
  - Djankov et al. (2010), Hummels and Schaur (2013), Alessandria et al. (2023), Carreras-Valle and Ferrari (2025)

More

Data

Input-output network

- BEA use table for 2017: 402 industries, 371 sectors kept

Measuring delays

$$\text{backlog ratio} = \frac{\text{stock value of unfilled orders}}{\text{flow value of goods delivered per month}}$$

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1. Census M3 survey: 7 durable-goods industries, each ratio applied to all its sectors: 124 sectors
2. Compustat order backlog: listed firms, but firm level and many sectors outside manufacturing: 135 sectors
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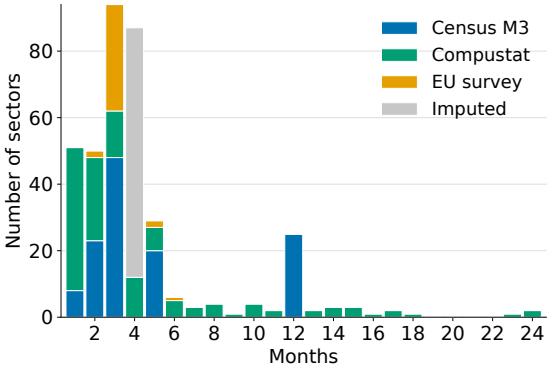
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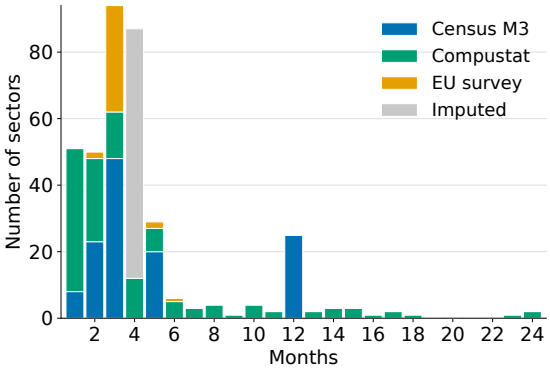
Later on, we show that this measure has bite in the data.

Backlog Distribution



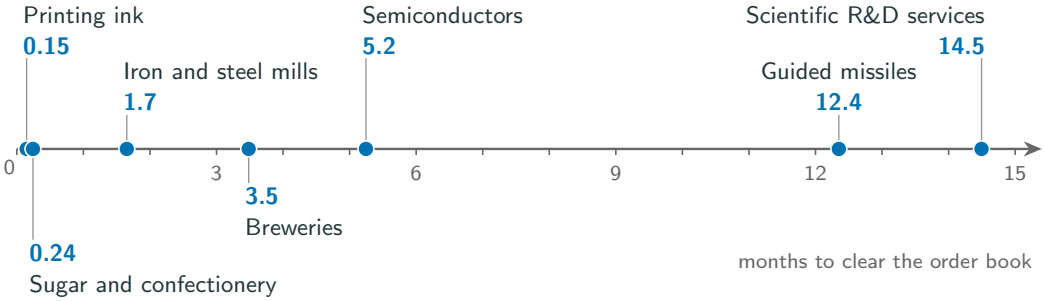
- Average 4.4 months, 5.0 weighted by value added

Backlog Distribution



- Average 4.4 months, 5.0 weighted by value added
- Measured for 296 sectors (53.5% of value added)
- The rest (gray), mostly agriculture and services, get the average of measured manufacturing, 4 months

Delays range from days to more than a year



Backlog ratios in months, before rounding

The whole network

Model

Model

- Time is discrete
- Representative household with inelastic labor supply, discount factor β
- Sectors $n = 1, \ldots, N$ with production

$$y_{nt} = A_{nt} F_n \left(l_{nt}, x_{n1,t}, \ldots, x_{nN,t} \right)$$

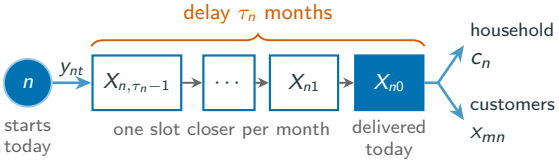
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  where A_{nt} is productivity, l_{nt} labor, $x_{nm,t}$ input of good m
- Time-to-build modeled as delivery lags:
  - Goods of sector n take τ_n periods to be delivered (the *delay* of n)
  - $X_{n\tau} \equiv$ stock of good n scheduled for delivery in τ periods
- No capital, no storage between periods



Planning Problem

Each period the planner sees the productivities $\{A_n\}$ and the goods in the pipeline $\{X_{n\tau}\}$, the state, and chooses consumption, labor and inputs:

$$V\left(\{A_n\}, \{X_{1\tau}\}_{\tau=0}^{\tau_1-1}, \ldots, \{X_{N\tau}\}_{\tau=0}^{\tau_N-1}\right) = \max_{c_n, l_n, x_{nm}, y_n} U\left(c_1, \ldots, c_N\right) + \beta E\left[V\left(\{A_n'\}, \{X_{1\tau}'\}_{\tau=0}^{\tau_1-1}, \ldots, \{X_{N\tau}'\}_{\tau=0}^{\tau_N-1}\right)\right]$$

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subject to the labor constraint $\sum_n l_n \leq 1$ and, for all $n = 1, \ldots, N$:

$X'_{n\tau} = X_{n,\tau+1}$ for $0 \leq \tau < \tau_n - 1$	goods in the pipeline move one period closer
$X'_{n,\tau_n-1} = y_n$	output started today is delivered in τ_n periods
$X_{n0} \geq c_n + \sum_m x_{mn}$	goods delivered today are consumed or used as inputs
$y_n = A_n F_n(l_n, x_{n1}, \ldots, x_{nN})$	

$\Rightarrow$ The state is the whole pipeline, 371 sectors $\times$ their lags: intractable in general

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Solution

- Way out: log utility and Cobb-Douglas production with constant returns

$$U = \log\left(\prod_n c_n^{\gamma_n}\right), \qquad F_n(l, x_1, \ldots, x_N) = l^{\alpha_n} \prod_m x_m^{\omega_{nm}}, \quad \alpha_n + \sum_m \omega_{nm} = 1$$

- Income and substitution effects cancel: the planner keeps labor and the use of each delivered good in constant shares

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Proposition

Under these assumptions $I_n = \overline{I}_n$, $c_n = \overline{c}_n X_{n0}$, $x_{mn} = \overline{x}_{mn} X_{n0}$ *(constants), and the value function is*

$$V = \sum_{n=1}^{N}\sum_{\tau=0}^{\tau_n - 1} \beta^\tau \zeta_n \log X_{n\tau} + G\left(\{A_n\}\right),$$

where $\zeta_n = \gamma_n + \sum_m \omega_{mn}\beta^{\tau_m}\zeta_m$ *and* $G\left(\{A_n\}\right) = \sum_n \beta^{\tau_n}\zeta_n \log A_n + \beta E\left[G\left(\{A'_n\}\right)\right]$.

- V = value of the goods in the pipeline(ζ) + value of productivity(G)

What ζ_n and G capture

- ζ_n = value of one more percent of good n, *once delivered*:

$$\zeta_n = \underbrace{\gamma_n}_{\substack{\text{consumption} \\ \text{of } n}} + \sum_m \underbrace{\omega_{mn}}_{\substack{\text{output of customer } m \\ \text{rises by } \omega_{mn}\%}} \underbrace{\beta^{\tau_m}}_{\substack{\text{delivered} \\ \tau_m \text{ periods later}}} \underbrace{\zeta_m}_{\substack{\text{worth } \zeta_m \\ \text{once delivered}}}$$

- Iterating: ζ_n adds up every supply chain from n to the household, weighted by its input shares and discounted by the delays along the way
  - Without delays, $\boldsymbol{\zeta} = (\mathbf{I} - \boldsymbol{\Omega}')^{-1}\boldsymbol{\gamma}$: the Domar weights of the static network

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  - Without delays, $\boldsymbol{\zeta} = (\mathbf{I} - \boldsymbol{\Omega}')^{-1}\boldsymbol{\gamma}$: the Domar weights of the static network
- G = value of productivity:
  - With i.i.d. shocks, $G = \sum_n \beta^{\tau_n}\zeta_n \log A_n$: a 1% shock to n is worth $\beta^{\tau_n}\zeta_n$
  - the extra output is delivered τ_n periods later (β^{τ_n}), and is then worth ζ_n
  - in the equilibrium $\beta^{\tau_n}\zeta_n$ is the *forward value of production*

- Decentralization: prices, GDP, and Hulten's theorem in the backup slides

Prices, GDP and Hulten | Aggregates

Output

- In log-deviation from steady state:

$$\hat{y}_{nt} = \hat{A}_{nt} + \sum_m \omega_{nm} \hat{y}_{m,t-\tau_m}$$

each input enters with its supplier's delay τ_m

- VAR(τ_{max}) representation:

$$\hat{\mathbf{y}}_t = \mathbf{\Omega}_1 \hat{\mathbf{y}}_{t-1} + \ldots + \mathbf{\Omega}_{\tau_{max}} \hat{\mathbf{y}}_{t-\tau_{max}} + \hat{\mathbf{A}}_t, \qquad \mathbf{\Omega}_\tau = \mathbf{\Omega}\, \mathbf{diag}\left(\mathbf{1}\{\tau = \tau_m\}\right)$$

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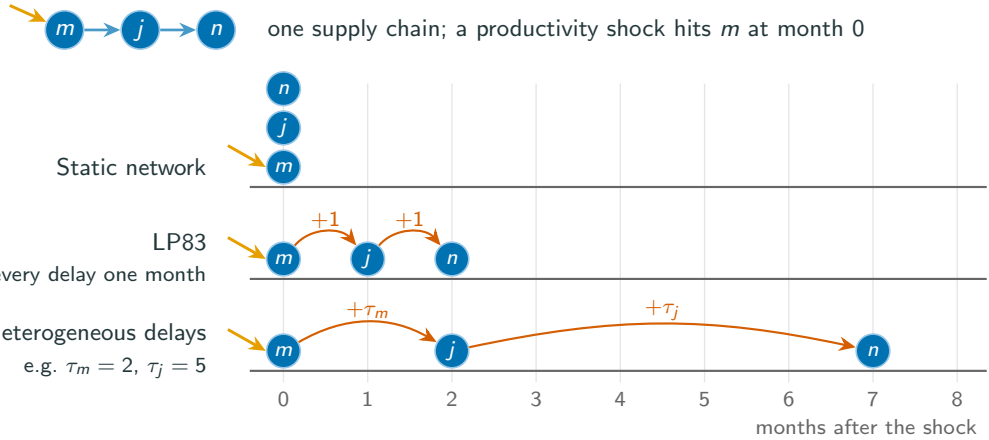
- Nested cases:
  - Roundabout production (static network): no time-to-build

$$\hat{\mathbf{y}}_t = \hat{\mathbf{A}}_t + \mathbf{\Omega}\hat{\mathbf{y}}_t \;\Rightarrow\; \hat{\mathbf{y}}_t = \hat{\mathbf{A}}_t + \mathbf{\Omega}\hat{\mathbf{A}}_t + \mathbf{\Omega}^2\hat{\mathbf{A}}_t + \ldots = [\mathbf{I} - \mathbf{\Omega}]^{-1}\hat{\mathbf{A}}_t \quad \text{(Leontief inverse)}$$

  - Long and Plosser (1983), *LP83*: one-period time-to-build

$$\hat{\mathbf{y}}_t = \hat{\mathbf{A}}_t + \mathbf{\Omega}\hat{\mathbf{y}}_{t-1} \;\Rightarrow\; \hat{\mathbf{y}}_t = \hat{\mathbf{A}}_t + \mathbf{\Omega}\hat{\mathbf{A}}_{t-1} + \mathbf{\Omega}^2\hat{\mathbf{A}}_{t-2} + \ldots$$

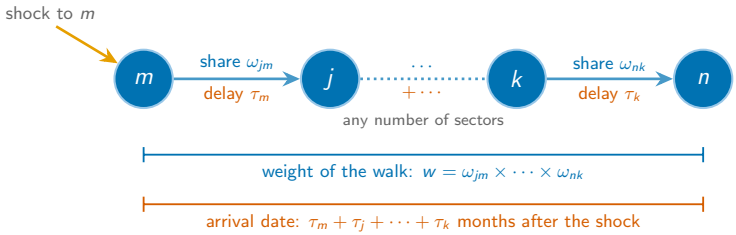
Same shock, three timings



⇒ Same chain in all three economies; the delays set only when each sector responds

Propagation along walks

- *Walk*: a supply chain of any length, $m \to j \to \cdots \to k \to n$
  - its *weight*: the product of the input shares along it
  - its *arrival date*: the sum of the delays along it, the last sector excepted



- *Cycle*: a walk from sector m to itself
  - its *weight*: same
  - its *duration*: total duration $\tau_m + \tau_j + ... + \tau_k$ to go back to itself

Persistence and Delays

- Persistence statistics

$$\mathcal{T}\left(\hat{\mathbf{A}}\right) = \frac{1}{CIR\left(\hat{\mathbf{A}}\right)} \sum_{\tau=0}^{\infty} \sum_{i} \tau w_i \hat{y}_{i\tau}\left(\hat{\mathbf{A}}\right)$$

  $\Rightarrow$ Substantial additional persistence even for iid shocks, highly heterogeneous

- Delay shocks
  - Delayed arrival of intermediate goods to later date

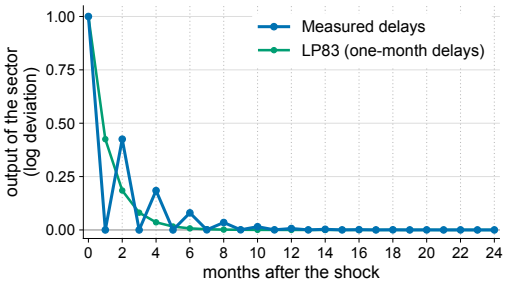
  $\Rightarrow$ Sizeable impact of delays, some leading to oscillations

- Bottlenecks
  - To which sectors do delay shocks contribute most to the persistence of shocks?

  $\Rightarrow$ Bottlenecks identified by (weighted) supplier $\times$ buyer centrality measures

Echoes and Endogenous Fluctuations

Nonferrous metal smelting echoes every two months



Output after a shock to the sector's own productivity

- Period: 2 months, the sector's delay
- Each wave is about 43% of the previous one = the sector's input share of its own output
- LP83: smooth decline, same total

Other sectors

- *Self-loop*: a sector that uses its own output, the simplest cycle
- Recall: a cycle's *duration* = sum of the delays along it; its *weight* = product of the input shares along it

$\Rightarrow$ Period = duration of the self-loop; ratio of successive waves = its weight

Does the whole economy behave like this sector? The eigenvalues

- Write the VAR(τ_{max}) as a VAR(1) by stacking the lags:

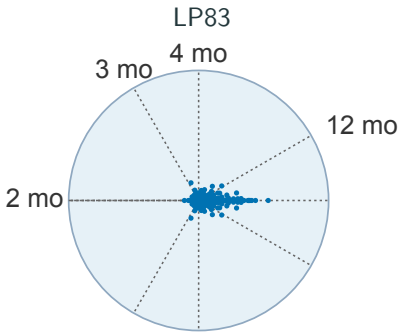
$$\underbrace{\begin{pmatrix} \hat{\mathbf{y}}_t \\ \hat{\mathbf{y}}_{t-1} \\ \vdots \\ \hat{\mathbf{y}}_{t-\tau_{max}+1} \end{pmatrix}}_{\equiv \mathbf{Y}_t} = \underbrace{\begin{pmatrix} \mathbf{\Omega}_1 & \mathbf{\Omega}_2 & \cdots & \mathbf{\Omega}_{\tau_{max}} \\ \mathbf{I} & \mathbf{0} & \cdots & \mathbf{0} \\ & \ddots & \ddots & \vdots \\ \mathbf{0} & \cdots & \mathbf{I} & \mathbf{0} \end{pmatrix}}_{\equiv \mathbf{O}} \underbrace{\begin{pmatrix} \hat{\mathbf{y}}_{t-1} \\ \hat{\mathbf{y}}_{t-2} \\ \vdots \\ \hat{\mathbf{y}}_{t-\tau_{max}} \end{pmatrix}}_{\equiv \mathbf{Y}_{t-1}} + \underbrace{\begin{pmatrix} \hat{\mathbf{A}}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{pmatrix}}_{\equiv \mathbf{e}_t}$$

- The system oscillates if $\mathbf{O}$ has complex eigenvalues
  - Only with time-to-build (NOT static model)!
- For each eigenvalue $\xi = |\xi| e^{i\theta}$:
  - Modulus $|\xi| < 1$ governs the decay
  - Argument $\theta \neq 0$ leads oscillations of period $2\pi/\theta$ months

The eigenvalues of O

Each dot: one eigenvalue. Distance to the center = modulus; rays = periods $2\pi/\theta$.

Histogram

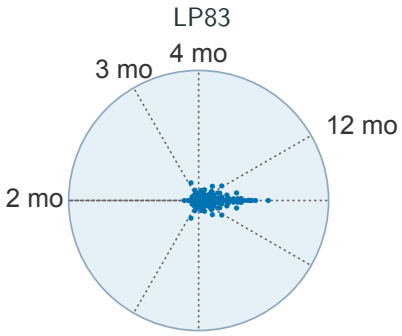


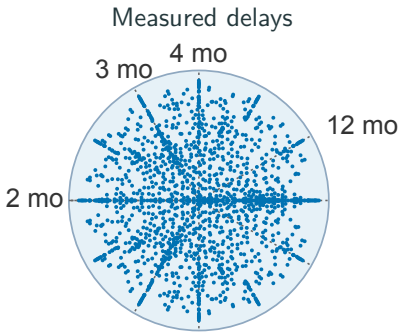
- LP83: small, near the positive real axis: shocks die out quickly and smoothly

The eigenvalues of O

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Histogram





- LP83: small, near the positive real axis: shocks die out quickly and smoothly
- Measured delays: larger (up to 0.92), along rays at periods of 2, 3, 4 and 12 months

$\Rightarrow$ Same input-output matrix in both. Where do the oscillations come from?

Oscillations and Network Cycles

- Oscillations are a consequence of cycles (loops) in the network
- A simple result:

Proposition (Acyclical network)

A vertical production network (i.e. acyclical) displays no oscillations.

Proof.

- There exists an ordering of sectors in which Ω is lower triangular with 0s on the diagonal
- All eigenvalues are 0
- Note: shocks vanish after a finite number of iterations (at most $N \times \tau_{max}$)

- Eigenvalues in the general case are too complicated
  - Algebraic graph theory: at most characterize 1st and 2nd largest eigenvalues...
  - ... but we can characterize the **Fourier spectrum**!

Refresher: Discrete-Time Fourier Transform (DTFT)

- Any discrete-time 0-mean stationary process x_t can be represented by

$$x_t = \int_{-\pi}^{\pi} \delta(\omega) e^{i\omega t} d\omega$$

  where $E[\delta(\omega)] = 0$, $E[\delta(\omega)\delta(\omega')] = 0$ for $\omega \neq \omega'$

- The Discrete Time Fourier Transform (DTFT) is

$$\delta(\omega) = \frac{1}{2\pi} \sum_{t=-\infty}^{\infty} x_t e^{-i\omega t}$$

- The spectral density is

$$f(\omega) \equiv E\left[\delta(\omega)\overline{\delta(\omega)}\right]$$

DTFT and Autocorrelation Function

- Autocorrelation function (ACF)

$$\gamma_k = E\left[x_t x_{t-k}\right] \text{ for } k = -\infty, ..., \infty$$

- **Key property**: Fourier spectrum is the DTFT of the ACF

$$f(\omega) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma_k e^{-i\omega k}$$

$\Rightarrow$ The ACF can be characterized analytically & using network topology

ACMF of a VAR(1)

- Recall the VAR(1) representation

$$\mathbf{Y}_t = \mathbf{O}\mathbf{Y}_{t-1} + \mathbf{e}_t$$

  and $\mathbf{\Sigma} = E[\mathbf{e}\mathbf{e}']$ and $\mathbf{e}$ *iid*

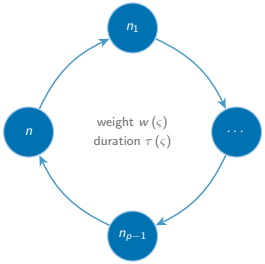
- The Autocovariance Matrix Function $\mathbf{\Gamma}_k = E\left[\mathbf{Y}_t\mathbf{Y}'_{t-k}\right]$ is

$$\mathbf{\Gamma}_0 = \sum_{k=0}^{\infty} \mathbf{O}^k \mathbf{\Sigma} \left(\mathbf{O}'\right)^k$$
$$\mathbf{\Gamma}_k = \mathbf{O}^k \mathbf{\Gamma}_0$$

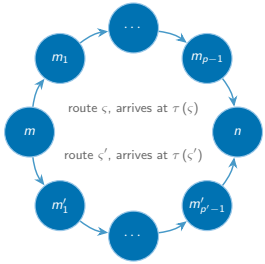
- We can extract the relevant $\gamma_k(i) = E[\hat{y}_{it}\hat{y}_{it-k}]$ and construct spectrum
  - ... but provides little understanding

Where serial autocorrelation comes from

- i.i.d. shocks: $\hat{y}_{nt}$ and $\hat{y}_{n,t-k}$ covary only if the same shock reaches n at both dates
- So: two walks from the shocked sector to n, arriving k months apart



(a) **Directed cycle**: same route, extra laps
Correlation at every multiple of the duration
(also for sectors downstream of the cycle)



(b) **Undirected cycle**: two routes
Correlation at one lag, $|\tau(\varsigma) - \tau(\varsigma')|$

$\Rightarrow$ Only directed cycles make serial correlation recur

A cycle leaves a comb in the autocovariances

- A cycle of weight w and duration τ through sector n
- After k laps, n's output contains w^k times its own value $k\tau$ months earlier
- The other pairs of walks only add to this (nonnegative terms)

Proposition (contribution of a cycle, i.i.d. shocks)

For a cycle of weight w and duration τ through sector n,

$$\gamma_{k\tau}(n) \geq w^{|k|}\sigma_n^2 \quad \text{for every } k, \qquad \sigma_n^2 = \operatorname{var}(\hat{y}_{nt})$$

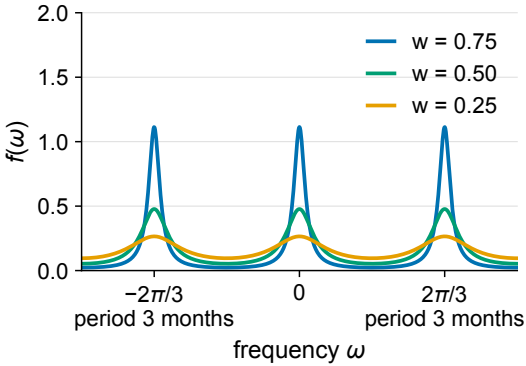
- A comb: a peak every τ months, the k-th at least $w^k\sigma_n^2$
- A sector on several cycles: their combs add up, still a lower bound

The same comb in the spectrum

Proposition (contribution of a cycle, continued)

If the cycle were the only source of serial correlation of sector n, its spectrum would be

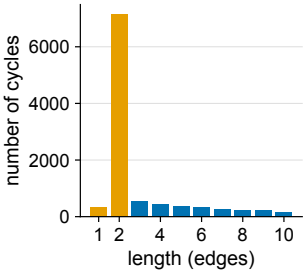
$$\frac{\sigma_n^2}{2\pi}\,\frac{1-w^2}{1+w^2-2w\cos(\omega\tau)}$$



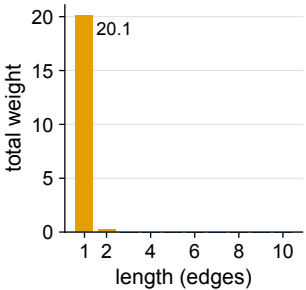
- Peaks at frequency $2\pi/\tau$ (period τ) and its multiples (fig: $\tau = 3$)
- Heavier cycle: sharper peaks
- Several cycles: the peaks of each add up

Harmonics Poisson

The heavy cycles of the U.S. network



(a) Cycles by length



(b) Total weight w, by length

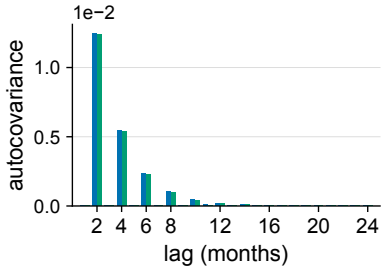
- Many cycles of any length
- By weight: self-loops and 2-cycles matter the most
- Largest: nonferrous metal 0.43, beef cattle 0.41, distilleries 0.39
- Two-sector cycles: at most 0.025
- Self-loop above 0.1, a strong echo: one sector in six (13% of output). Median 0.02

lengths 1 and 2, all listed; longer cycles, found by 250 random crawlers

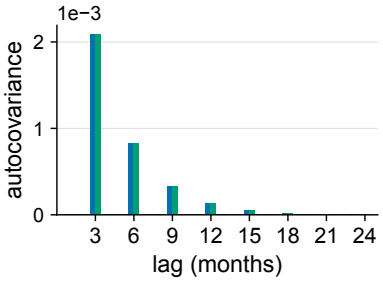
The cycles account for the serial correlation of the model

- The proposition is a lower bound. How tight?
- Per sector: sum $w^k \sigma_n^2$ over its directed cycles; compare with the model at lags 1–60
- Pooled $R^2 = 0.97$, median sector 0.85. Self-loops alone: the same

Lags 1 to 24 months: autocovariances of the model; sum of the directed-cycle contributions



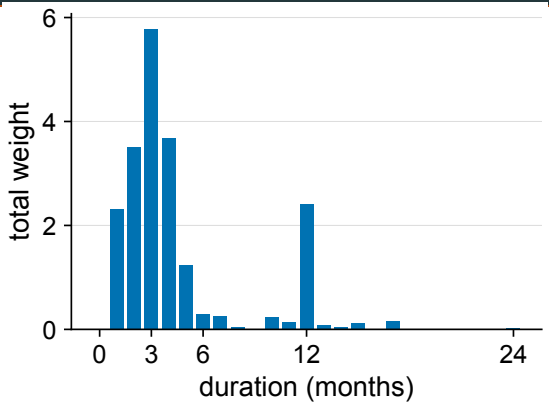
(a) Nonferrous metal: self-loop 0.43, 2 months



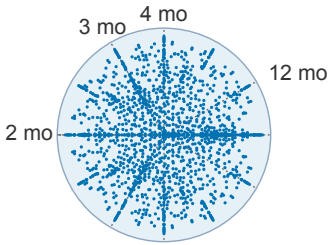
(b) Distilleries: self-loop 0.39, 3 months

$\Rightarrow$ The bound is nearly the whole story

The durations of the heavy cycles are the rays of the eigenvalues



Total cycle weight, by duration



Eigenvalues, measured delays

- Cycle weight at 2, 3, 4 and 12 months; eigenvalue rays at 2, 3, 4 and 12 months
- Caveat: 84% of the 4-month weight is sectors with no backlog data, set to 4
- LP83: self-loop lasts one month, others too small, so no rays

Sectoral Comovements and Aggregation

From sectors to aggregate output

- Aggregate output: $\hat{y}_t = \sum_n \mu_n \hat{y}_{nt}$, μ_n = value-added share (as in industrial production)
- Its serial correlation has two sources
  - each sector's own: the cycles of the previous section
  - comovement between sectors
- Comovement of n and m at lag l, with i.i.d. shocks:

$$[\mathbf{\Gamma}_l]_{nm} = \sum_j \sigma^2(\hat{A}_j) \sum_{\substack{\varsigma: j \to n,\ \varsigma': j \to m \\ \tau(\varsigma) - \tau(\varsigma') = l}} w(\varsigma)\, w(\varsigma')$$

- $\mathbf{\Gamma}_l = E[\hat{\mathbf{y}}_t \hat{\mathbf{y}}'_{t-l}]$: each pair of walks from a common source j, arriving l months apart, adds the product of its weights times the variance of the shock to j

$\Rightarrow$ Do the echoes survive aggregation?

How we compute it | Real GDP and the other aggregates

Two sources of aggregate fluctuations

Proposition

The spectrum of aggregate output is

$$f_y(\omega) = \underbrace{\sum_{n=1}^{N} \mu_n^2 f_n(\omega)}_{\text{sum of sectoral spectra}} + \underbrace{\frac{1}{2\pi} \sum_{n \neq m} \sum_{k=-\infty}^{\infty} \mu_n \mu_m \left[\mathbf{\Gamma}_k\right]_{nm} e^{-i\omega k}}_{\text{sectoral comovement term}}$$

with $\mathbf{\Gamma}_k = E[\hat{\mathbf{y}}_t \hat{\mathbf{y}}'_{t-k}]$ *the autocovariances of sectoral output.*

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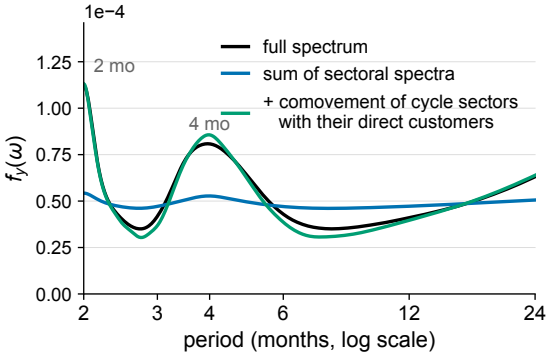
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- Sectoral spectra: each has peaks at its cycles' durations; 371 small weights mute them, but cannot cancel them
- Comovement: pairs of walks from a common shock
  - a cycle sector and its customers echo together, at the cycle's duration
  - so this term peaks at the same periods and brings them back

Transfer form

The echoes survive aggregation



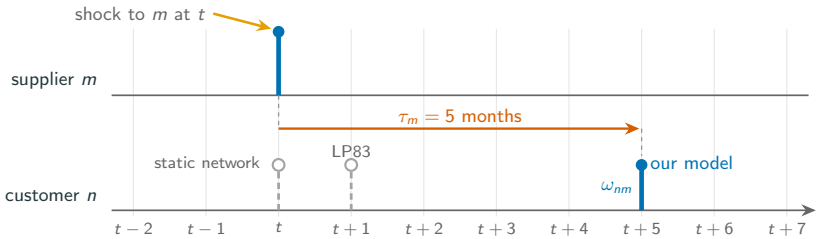
- 371 sectors with small weights could average the echoes away
- They do not: peaks at 4 and 2 months many self-loops
- Each sector: its directed cycles explain its own serial correlation
- The cycles' echoes reach the aggregate through their customers, who echo with them (green)

The 4-month peak partly reflects the default delay of sectors without backlog data.

What the delays do to the business cycle

Empirical Evidence

A supplier's shock reaches its customers exactly one delay later



Law of motion

$$\hat{y}_{nt} = \hat{A}_{nt} + \sum_m \omega_{nm}\, \hat{y}_{m,t-\tau_m}$$

In growth rates

$$\Delta\hat{y}_{nt} = \Delta\hat{A}_{nt} + \underbrace{\sum_m \omega_{nm}\, \Delta\hat{y}_{m,t-\tau_m}}_{X_{nt}^{(\tau)},\ \text{the delayed composite}}$$

- No free parameter: the shares, the delays and the outputs are all measured

Bringing the model to the data

- We need high-frequency (monthly) output (quantity) by industry
  - Fed's industrial prod. indexes (G.17), 104 ind., 1972–2026, matched to BEA sectors
- Input shares ω from the BEA tables, delays τ from the backlog data.
- Two things the G.17 does not give us
  - all suppliers: it covers manuf., mining and utilities, about 2/3 of inputs for typical ind.
  - a clean date for output: some indexes record output when it is built, others when it is sold

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  - a clean date for output: some indexes record output when it is built, others when it is sold
- Our answer: simulate each economy and measure it exactly like the data
  - 100 panels per economy: same missing suppliers, same recording dates, same regression
  - An economy's prediction: the distribution of its 100 estimates

Pipeline

Two predictions from network cycles

- Self-loop echo: n's growth vs. its own growth τ_n months earlier; correlation of at least ω_{nn}
- Trading-pair arm: n and m sell to each other; n's growth vs. m's growth τ_m months earlier
- The echo is not automatic: a BEA "self-loop" can be two industries inside one sector buying from each other, with no echo. Finding it is informative

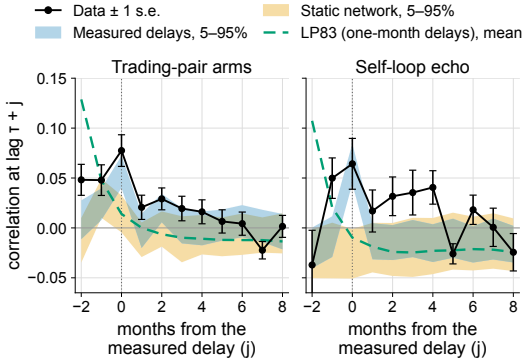
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How we measure

- Correlation of n's growth with the partner's growth at lag $\tau + j$, for $j = -2, \ldots, 8$; averaged over the pairs (or sectors)
- Prediction: a spike at $j = 0$. Static network: none. LP83: at lag one, that is at $j = 1 - \tau$, two or more months before the delay
- **Excess correlation**: value at $j = 0$ minus the average at $j = \pm 2, \pm 3$

The network's cycles show in the data at the measured delays



Excess correlation

	Trading-pair arms	Self-loop echo
Data	$+\mathbf{0.045}$	$\mathbf{0.053}$
(s.e.)	(0.013)	(0.022)
Our model	0.047	0.078
Static	$+0.02$	≈ 0
LP83	-0.03	-0.03

- Data: a spike at $j = 0$ in both panels, inside our model's band
- Static: flat. LP83: elevated at $j \leq -2$ (its lag one), negative at the delay

Bands: 5th–95th percentiles of 100 simulated panels. Echo: 29 industries with delays ≥ 3 months, indexes at least 80% built-dated. Arms: 53 pairs with input shares ≥ 0.002 both ways and both delays ≥ 3 months.

Conclusion

- Delays change when a shock arrives, not how much arrives: longer, uneven persistence across sectors
- The economy fluctuates at the durations of the network's cycles: echoes, mostly from self-loops, visible in aggregate output
- Evidence: in monthly industrial production, shocks reach customers at the measured delays, with no free parameter. The data favor the model over the static network and LP83

More in the paper

- The welfare cost of delays and the bottleneck sectors
- Measured delays damp aggregate fluctuations at business-cycle periods
- Comovement across sectors through pairs of walks

Persistence and Delays

Persistence: the average duration of a shock

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- Delays leave the total effect of a shock unchanged; they set *when* it arrives

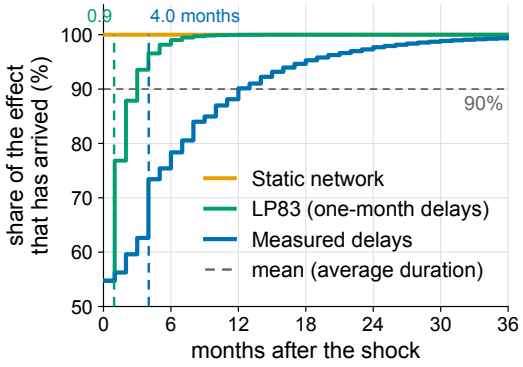
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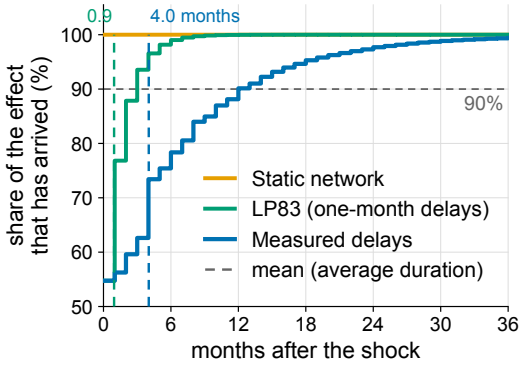
Shocks take several times longer to work through the economy



Unit shock to every sector: share of the effect arrived by month t

- Static network: all on impact
- LP83 and our model: 55% on impact (shocked sectors' own output)
- LP83: the rest within a few months
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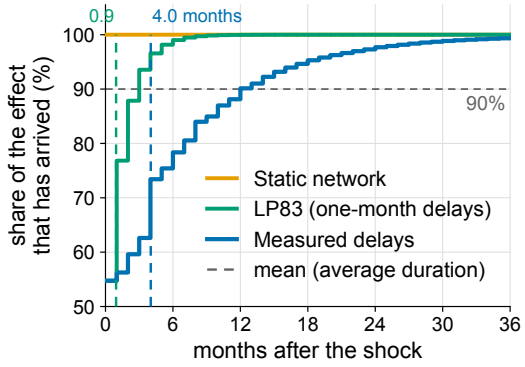
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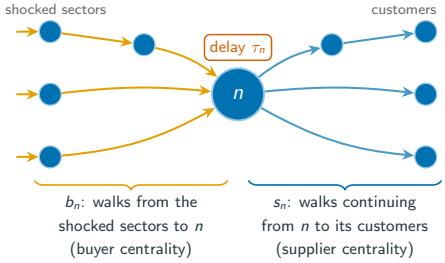
- Shocks to a single sector: a fifth of the sectors take 7 months or more (none above 3 in LP83); guided missiles, 32.7 months

Average duration = delays × traffic

- Each walk carries part of the total effect. $\mathcal{T}$ is the average arrival date across walks, each weighted by its part
- A walk is delayed τ_n each time it passes through n (the last sector excepted), so

$$\mathcal{T} = \sum_n \tau_n \, v_n$$

- v_n: the traffic through n, the average number of passages through n
- To compute it, we count the walks through n: those arriving at n, and those leaving it



Two centralities: walks into n and walks out of n

- $[\mathbf{\Omega}^k]_{nm}$: total weight of the walks of k steps from m to n
- Buyer centrality b_n: every walk from a shocked sector to n

$$b_n = \sum_m \sum_{\varsigma:\, m \to n} w(\varsigma)\, \hat{A}_m = \left[(\mathbf{I} + \mathbf{\Omega} + \mathbf{\Omega}^2 + \ldots)\hat{\mathbf{A}}\right]_n = \left[(\mathbf{I} - \mathbf{\Omega})^{-1}\hat{\mathbf{A}}\right]_n$$

  - the cumulative response of n's output to the shock
  - large if n buys, directly or indirectly, from the shocked sectors

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  - the cumulative response of n's output to the shock
  - large if n buys, directly or indirectly, from the shocked sectors
- Supplier centrality s_n: every walk of at least one step from n to a sector j, weighted by j's size μ_j

$$s_n = \sum_j \mu_j \sum_{\varsigma:\, n \to j} w(\varsigma) = \left[\boldsymbol{\mu}'(\mathbf{\Omega} + \mathbf{\Omega}^2 + \ldots)\right]_n = \left[\boldsymbol{\mu}'\mathbf{\Omega}(\mathbf{I} - \mathbf{\Omega})^{-1}\right]_n$$

  - the cumulative response of aggregate output to a unit shock to n, net of n's impact
  - large if n supplies, directly or indirectly, large sectors

The traffic through n is the product of the two

- A walk through n = a walk into n, followed by a walk out of n
- So $b_n \times s_n$ adds up every walk through n, once per passage, weighted by the shock at its start and the size of its end
- Divided by the total effect (CIR, the same sum over *all* walks): the expected number of passages through n

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- A sector that receives much of the shock but supplies no one ($s_n = 0$) has zero traffic: lengthening its delay postpones nothing

Proof

The traffic does not depend on the delays

- Neither b_n nor s_n involves the delays: the network sets which walks a shock takes and how much each carries; the delays set only the arrival dates
- So the average duration is *linear* in the delays, with the traffic as coefficients:

$$\mathcal{T} = \sum_n \tau_n \, v_n, \qquad \frac{\partial \mathcal{T}}{\partial \tau_n} = v_n$$

- A duration is long when walks pass through long delays (guided missiles), or pass through the same sectors many times (cycles: the effect comes back in echoes, next section)

$\Rightarrow$ One more month of delay in sector n adds v_n months to the average duration

Appendix: bottlenecks (whose delays matter the most), welfare cost of the delays Bottlenecks Welfare

Persistence and Delays

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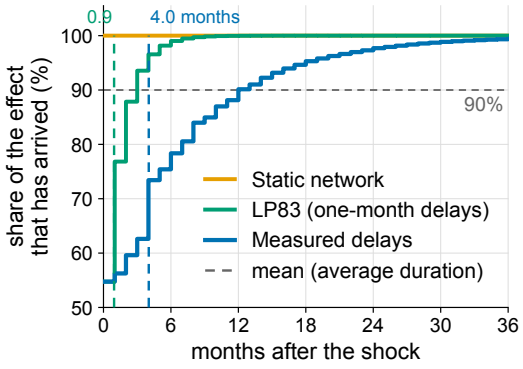
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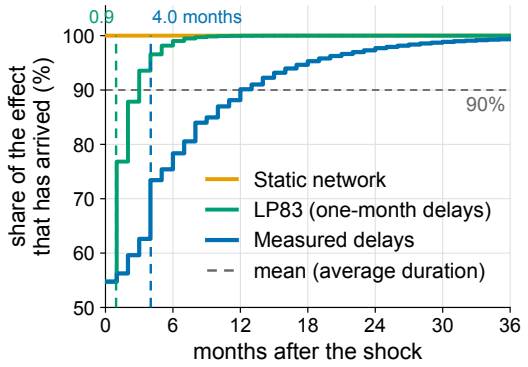
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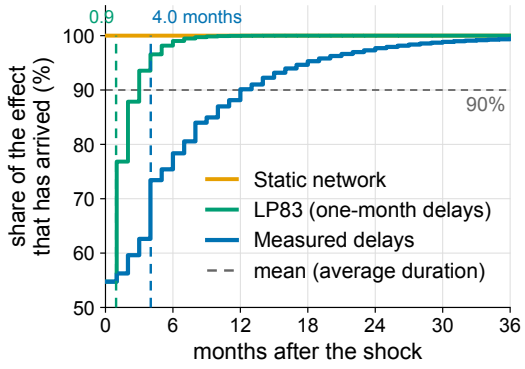
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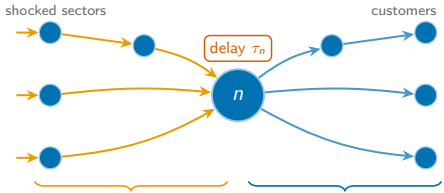
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- $[\mathbf{\Omega}^k]_{nm}$: total weight of the walks of k steps from m to n
- Buyer centrality b_n: every walk from a shocked sector to n

$$b_n = \sum_m \sum_{\varsigma:\, m \to n} w(\varsigma)\, \hat{A}_m = \left[(\mathbf{I} + \mathbf{\Omega} + \mathbf{\Omega}^2 + \ldots)\hat{\mathbf{A}}\right]_n = \left[(\mathbf{I} - \mathbf{\Omega})^{-1}\hat{\mathbf{A}}\right]_n$$

  - the cumulative response of n's output to the shock
  - large if n buys, directly or indirectly, from the shocked sectors

- Supplier centrality s_n: every walk of at least one step from n to a sector j, weighted by j's size μ_j

$$s_n = \sum_j \mu_j \sum_{\varsigma:\, n \to j} w(\varsigma) = \left[\boldsymbol{\mu}'(\mathbf{\Omega} + \mathbf{\Omega}^2 + \ldots)\right]_n = \left[\boldsymbol{\mu}'\mathbf{\Omega}(\mathbf{I} - \mathbf{\Omega})^{-1}\right]_n$$

  - the cumulative response of aggregate output to a unit shock to n, net of n's impact
  - large if n supplies, directly or indirectly, large sectors

The traffic through n is the product of the two

- A walk through n = a walk into n, followed by a walk out of n
- So $b_n \times s_n$ adds up every walk through n, once per passage, weighted by the shock at its start and the size of its end
- Divided by the total effect (CIR, the same sum over *all* walks): the expected number of passages through n

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Proposition (average duration)

$$\mathcal{T} = \sum_n \tau_n \, v_n, \qquad v_n = \frac{b_n \times s_n}{\mathrm{CIR}}$$

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- A sector that receives much of the shock but supplies no one ($s_n = 0$) has zero traffic: lengthening its delay postpones nothing

Proof

The traffic does not depend on the delays

- Neither b_n nor s_n involves the delays: the network sets which walks a shock takes and how much each carries; the delays set only the arrival dates
- So the average duration is *linear* in the delays, with the traffic as coefficients:

$$\mathcal{T} = \sum_n \tau_n \, v_n, \qquad \frac{\partial \mathcal{T}}{\partial \tau_n} = v_n$$

- A duration is long when walks pass through long delays (guided missiles), or pass through the same sectors many times (cycles: the effect comes back in echoes, next section)

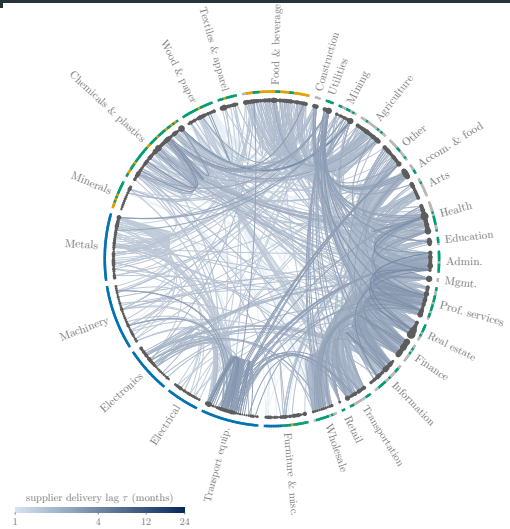
$\Rightarrow$ One more month of delay in sector n adds v_n months to the average duration

Appendix: bottlenecks (whose delays matter the most), welfare cost of the delays

Bottlenecks Welfare

Backup

The network and its delays



- Each point is a sector, sized by its Domar weight (its sales over GDP)
- Chords: the 1,500 largest input flows, darker when the supplier's delay is longer
- Outer ring: the source of each sector's delay, colored as in the histogram
- Long delays: durable manufacturing, and project services (engineering, research, computer-systems design)
- Both ship their output throughout the economy

Back

Prices: two markets, one at each end of production

- Prices are the planner's shadow values, in units of the household's utility
- A firm buys its inputs on the **spot** market, for delivery today:

$$p_{nt} = \frac{\partial V}{\partial X_{n0,t}} = \frac{\zeta_n}{X_{n0,t}} \qquad \text{value of one more unit of } n \text{ delivered today}$$

- It sells its output **forward**, for delivery τ_n periods later:

$$q_{nt} = \beta E_t \left[\frac{\partial V}{\partial X_{n,\tau_n - 1,\, t+1}} \right] = \beta^{\tau_n} \frac{\zeta_n}{y_{nt}} \qquad \text{value today of that future delivery}$$

  - in steady state $q_n = \beta^{\tau_n} p_n$: the spot price, discounted over the wait
- Every nominal magnitude is constant: spot value of the delivered stock $p_{nt} X_{n0,t} = \zeta_n$; forward revenue $q_{nt} y_{nt} = \beta^{\tau_n} \zeta_n$; spending on input m = the share ω_{nm} of revenue; consumption expenditure $\sum_n p_{nt} c_{nt} = \sum_n \gamma_n = 1$

$\Rightarrow$ Forward revenue $\beta^{\tau_n} \zeta_n$ is the marginal social value of a shock to n

Next: value added and GDP ◂ Back

Value added, nominal GDP and Domar weights

- Value is created at **production**, when the inputs are used: the national-accounts convention (goods in process count as output when produced)
- Value added = forward revenue − inputs bought at spot prices. Inputs absorb the share $1-\alpha_n$ of revenue, so

$$\mathrm{VA}_n = q_{nt}y_{nt} - \sum_m p_{mt}x_{nm,t} = \alpha_n \beta^{\tau_n}\zeta_n \qquad = \text{the wage bill of } n\text{: labor is the only primary factor}$$

- Nominal GDP = sum of value added, a constant: $\mathrm{GDP}^{\mathrm{nom}} = \sum_n \alpha_n \beta^{\tau_n}\zeta_n$
- **Domar weight** = forward sales over nominal GDP. Delays enter only through β^{τ_n}; without discounting it is the static network's Domar weight:

$$\lambda_n = \frac{q_{nt}y_{nt}}{\mathrm{GDP}^{\mathrm{nom}}} = \frac{\beta^{\tau_n}\zeta_n}{\sum_k \alpha_k \beta^{\tau_k}\zeta_k} \xrightarrow{\beta\to 1} \lambda_n^0 = \left[(\mathbf{I}-\mathbf{\Omega}')^{-1}\boldsymbol{\gamma}\right]_n$$

Next: real GDP and Hulten ◂ Back

Real GDP and Hulten's theorem

Definition. Real GDP is the chained (Divisia) volume index of value added: double-deflated sectoral real value added, aggregated with value-added shares (constant here, so no chaining is needed):

$$d\log Q_{nt}^{\mathrm{VA}} = \frac{q_{nt}y_{nt}}{\mathrm{VA}_{nt}}\, d\log y_{nt} - \sum_m \frac{p_{mt}x_{nm,t}}{\mathrm{VA}_{nt}}\, d\log x_{nm,t}, \qquad d\log Y_t = \sum_n \frac{\mathrm{VA}_{nt}}{\mathrm{GDP}^{\mathrm{nom}}}\, d\log Q_{nt}^{\mathrm{VA}}$$

Proposition

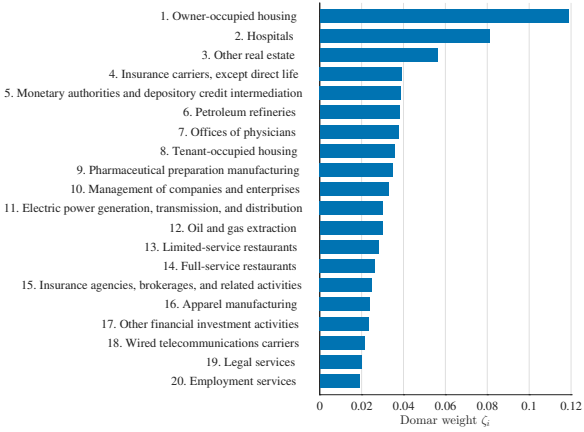
Real GDP growth is the Domar-weighted sum of productivity growth:

$$d\log Y_t = \sum_n \lambda_n\, d\log A_{nt}$$

- Hulten's theorem survives heterogeneous time-to-build. With i.i.d. shocks it also measures welfare: $dV = \sum_n q_{nt}y_{nt}\, d\log A_{nt} = \mathrm{GDP}^{\mathrm{nom}} \times d\log Y_t$ (a 1% shock to n adds 1% of its output, worth its forward price)
- Real GDP is **blind to the delays**: value added is counted at production, so the extra output counts today, as goods in process; with i.i.d. shocks it is white noise

Aggregates ◂ Back

Domar Weights



Top 20 sectors by ζ_n, the value of a delivered good (spot sales; the axis label). The model's Domar weight discounts it for the wait: $\lambda_n = \beta^{\tau_n}\zeta_n/\mathrm{GDP}^{\mathrm{nom}}$.

Prices and GDP

Back

Three aggregates see a shock at different times

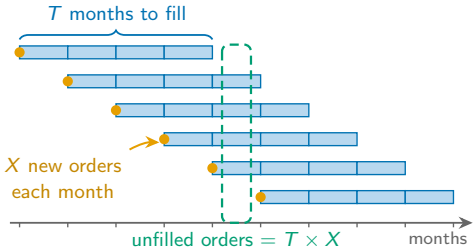
With $\mathbf{M}(z) = (\mathbf{I} - \boldsymbol{\Omega}\,\mathbf{diag}(z^{\boldsymbol{\tau}}))^{-1}$, the sum of $w(\varsigma)z^{\tau(\varsigma)}$ over walks, output is $\hat{\mathbf{y}}_t = \mathbf{M}(L)\hat{\mathbf{A}}_t$.

- **Real GDP** (chained index of value added): $d \log Y_t = \sum_n \lambda_n d \log A_{nt}$. With i.i.d. shocks it is white noise, because value added is counted at production. It is blind to the delays
- **Consumption**: $\hat{C}_t = \boldsymbol{\gamma}'\mathbf{M}^D(L)\hat{\mathbf{A}}_t$, $\mathbf{M}^D(z) = \mathbf{diag}(z^{\boldsymbol{\tau}})\mathbf{M}(z)$: one more delivery leg, so nothing on impact; its discounted sum is the welfare value of the shock
- **Aggregate output**: $\hat{y}_t = \boldsymbol{\mu}'\mathbf{M}(L)\hat{\mathbf{A}}_t$, $\mu_n = \mathrm{VA}_n/\mathrm{GDP}^{\mathrm{nom}} = \alpha_n\lambda_n$: value-added-weighted gross output, what industrial production measures. It carries the full propagation dynamics

$\Rightarrow$ In the static network all three move on impact; delays separate their timing

◂ Back: Solution ◂ Back: Aggregation

The backlog ratio measures the delay



- Backlog ratio = unfilled orders / monthly shipments = months needed to clear the order book. In the model's steady state it equals the delay exactly
- Measured for 296 of the 371 sectors, 53.5 percent of value added; the rest receive the average of the measured manufacturing sectors, 4 months

Back

Further related literature

- **Growth accounting with intermediate inputs**: Hulten (1978)
- **Distortions in production networks**: Jones (2011), Liu (2019), Baqaee and Farhi (2019)
- **Delays in supply chains**: Meier (2020)

Back

Average duration: differentiate the walk sum at $z = 1$

- Response of aggregate output $\sum_n \mu_n \hat{y}_{nt}$ to a shock $\hat{\mathbf{A}}$ at date 0, as a power series:

$$W(z) = \sum_{t \geq 0} z^t \sum_n \mu_n \hat{y}_{nt} = \boldsymbol{\mu}' \mathbf{M}(z) \hat{\mathbf{A}}, \qquad \text{CIR} = W(1), \qquad \mathcal{T} = \frac{W'(1)}{W(1)}$$

- $\mathbf{M}'(1) = \mathbf{M}(1)\, \boldsymbol{\Omega}\, \mathbf{diag}(\boldsymbol{\tau})\, \mathbf{M}(1)$ with $\mathbf{M}(1) = (\mathbf{I} - \boldsymbol{\Omega})^{-1}$. Writing $\mathbf{diag}(\boldsymbol{\tau}) = \sum_n \tau_n \boldsymbol{\delta}_n \boldsymbol{\delta}_n'$:

$$\mathcal{T} = \sum_n \tau_n \frac{s_n\, b_n}{\text{CIR}}, \qquad b_n = \boldsymbol{\delta}_n' (\mathbf{I} - \boldsymbol{\Omega})^{-1} \hat{\mathbf{A}}, \quad s_n = \boldsymbol{\mu}' \boldsymbol{\Omega} (\mathbf{I} - \boldsymbol{\Omega})^{-1} \boldsymbol{\delta}_n$$

- Walk reading: a walk arrives τ_n later for each passage through n (its last sector excluded). $b_n s_n$ adds every walk once per passage through n, so $v_n = s_n b_n / \text{CIR}$ is the expected number of passages: the traffic
- The traffic does not depend on the delays, so the duration is linear in them

Back

Bottlenecks: the sectors whose delays matter the most

- A supply disruption or a logistics failure adds months to one sector's delay. We ask which sectors slow the economy the most when that happens
- $\mathcal{T}$ is linear in the delays, with the traffic as coefficients:

Corollary (bottlenecks)

One more month in sector n's delay raises the average duration by the traffic through n: $\partial \mathcal{T} / \partial \tau_n = v_n = b_n s_n / \mathrm{CIR}$

- A bottleneck needs both factors: central as a buyer *and* as a supplier
- Largest traffic: other real estate, insurance agencies, management of companies, basic organic chemicals
- Size is not enough: owner-occupied housing and hospitals, the two largest sectors by Domar weight, rank 356th and 180th. They end supply chains: almost nothing passes through them to another sector
- The traffic does not depend on the delays, so the ranking does not use the delay data

Table Top 20 Back

Propagation bottlenecks are input hubs; welfare bottlenecks are large sectors

v_n, v_n^W: rise in the average duration $\mathcal{T}$ and the welfare duration $\mathcal{T}^W$ (unit shock to every sector) per extra month of delay in n; λ_n: Domar weight.

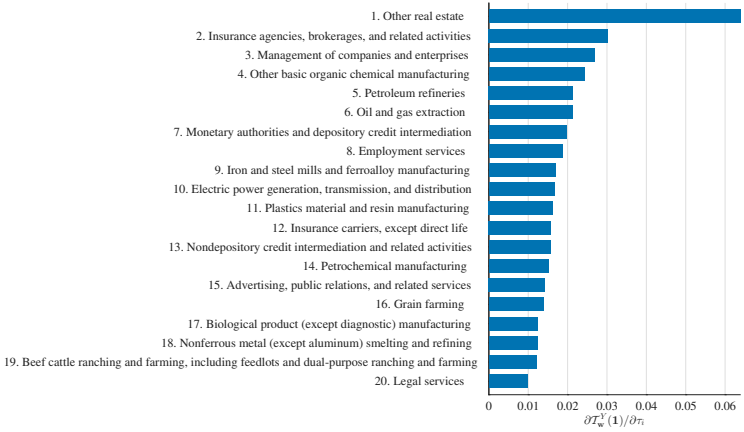
Propagation bottlenecks				Welfare bottlenecks			
Sector	v_n	λ_n	rank as welfare bottleneck	Sector	v_n^W	λ_n	rank as propagation bottleneck
Other real estate	0.064	0.057	3	Owner-occupied housing	0.079	0.120	356
Insurance agencies	0.030	0.025	16	Hospitals	0.078	0.082	180
Management of companies	0.027	0.033	10	Other real estate	0.066	0.057	1
Basic organic chemicals	0.024	0.013	22	Petroleum refineries	0.051	0.039	5
Petroleum refineries	0.021	0.039	4	Insurance carriers	0.039	0.040	12

- Propagation ($v_n \propto s_n b_n$): central buyer *and* supplier; size is not enough (housing, hospitals)
- Welfare ($v_n^W \propto \lambda_n^0 b_n$, the household counts as a customer): mostly size; 7 of the 8 largest sectors by Domar weight are in the top 8
- The traffic does not depend on the delays, so the ranking does not use the delay data

Top 20

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Bottlenecks



Top 20 propagation bottlenecks: rise in the average duration of aggregate output per extra month of delay in the sector, for a unit shock to every sector.

Table

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The welfare cost of delays: the welfare duration

- The household discounts the future: gains that arrive later are worth less
- The household consumes delivered goods: $\hat{c}_{nt} = \hat{y}_{n,t-\tau_n}$
- So the effect of a shock on consumption is its effect on output, delayed once more by the producing sector's own delay
- Welfare duration $\mathcal{T}^W$: average duration of the effect on consumption, with the consumption shares γ_n as weights
- Unit shock to every sector: 9.0 months, against 2.0 in LP83
  - longer than the durations of output (4.0 and 0.9): one more delay

The welfare cost of delays: the value of a shock

Proposition (welfare value of a shock)

$$\Delta V = \underbrace{\boldsymbol{\lambda}^{0\prime}\hat{\mathbf{A}}}_{\text{value in the static network}} \times \underbrace{E\left[\beta^{D}\right]}_{\text{discount over the wait}} = \boldsymbol{\lambda}^{0\prime}\hat{\mathbf{A}}\left(1 - r\,\mathcal{T}^{W} + O(r^{2})\right)$$

- $\boldsymbol{\lambda}^{0}$: Domar weights of the static network (Hulten's theorem: the first factor is the value of the shock without delays)
- D: the random month in which the effect reaches the household, $E[D] = \mathcal{T}^{W}$; r: monthly discount rate
- The delays cost the share $1 - E[\beta^{D}] \approx r\,\mathcal{T}^{W}$ of the static value: 2.9% at 4% a year, against two thirds of a percent in LP83
- Shock to guided missiles: welfare duration 56.5 months, 17% of the value lost (exact share; the first-order $r\,\mathcal{T}^{W}$ gives 18.5%)
- Welfare bottlenecks: the household counts as a customer, so $\partial\mathcal{T}^{W}/\partial\tau_{n} \propto \lambda^{0}_{n} b_{n}$: mostly size; housing and hospitals lead

Back

Autocovariances of the VAR(1)

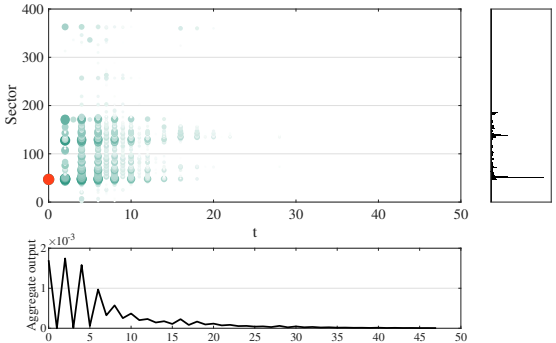
- $\mathbf{Y}_t = \mathbf{O}\mathbf{Y}_{t-1} + \mathbf{e}_t$ with $\mathbf{\Sigma}_e = E[\mathbf{e}_t \mathbf{e}_t']$ and $\mathbf{\Gamma}_k = E[\mathbf{Y}_t \mathbf{Y}_{t-k}']$:

$$\mathbf{\Gamma}_0 = \sum_{k=0}^{\infty} \mathbf{O}^k \mathbf{\Sigma}_e \left(\mathbf{O}'\right)^k, \qquad \mathbf{\Gamma}_k = \mathbf{O}^k \mathbf{\Gamma}_0$$

- The first $N \times N$ block of $\mathbf{\Gamma}_k$ is the sectoral $E[\hat{\mathbf{y}}_t \hat{\mathbf{y}}_{t-k}']$: its (n, m) entries are the comovements, its (n, n) entries the autocovariances $\gamma_k(n)$ of sector n
- This computes every sector's autocovariances and spectrum, but says nothing about where they come from: the cycles do

Back to echoes Back to aggregation

A shock to nonferrous metal smelting spreads in two-month waves



Shock of 1% to sector #331410 (red dot at $t = 0$). Top: response of each of the 371 sectors by month (dot size and shade = magnitude, log scale); right: cumulated response by sector. Bottom: aggregate output, in percent.

Back

Eigenvalues and cycles: the general statement

$\mathbf{O}$ = companion matrix of the VAR; a response is $\sum_j \xi_j^t c_j$ over its eigenvalues ξ_j. A first-return walk ς of n leaves n and comes back to it for the first time; the *first-return function* is $R_n(z) = \sum_\varsigma w(\varsigma) z^{\tau(\varsigma)}$.

Proposition

(i) All eigenvalues of $\mathbf{O}$ *are zero iff the network is vertical (has no cycle). (ii) For any sector n, every solution* ξ *of* $R_n(\xi^{-1}) = 1$ *is an eigenvalue of* $\mathbf{O}$*. (iii) An isolated cycle (its sectors lie on no other cycle) of weight* w *and duration* τ *contributes the* τ *roots* $\xi_k = w^{1/\tau} e^{2\pi i k/\tau}$, $k = 0, \ldots, \tau - 1$.

- Why (ii): after impact, $\hat{y}_{n,t} = \sum_\varsigma w(\varsigma) \hat{y}_{n,t-\tau(\varsigma)}$, and ξ^t solves it iff $R_n(\xi^{-1}) = 1$
- (iii) is the special case of (ii) with $R_n(z) = wz^\tau$: the cycle is n's only first-return walk
- U.S.: 345 of 371 sectors are connected by cycles; six self-loops are isolated

Back

Complex eigenvalues give a real response

- $\mathbf{O}$ is real, so its complex eigenvalues come in conjugate pairs: if $\mathbf{O}\mathbf{v} = \xi\mathbf{v}$, then $\mathbf{O}\bar{\mathbf{v}} = \bar{\xi}\bar{\mathbf{v}}$
- The shock is real, so the two pieces get conjugate sizes a and $\bar{a}$
- Their sum is real:

$$a\,\xi^t\mathbf{v} + \bar{a}\,\bar{\xi}^t\bar{\mathbf{v}} = 2\,\mathrm{Re}\left(a\,\xi^t\mathbf{v}\right)$$

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- For sector n, with $\xi = |\xi|e^{i\theta}$:

$$2\,|a\,v_n|\;|\xi|^t\cos\left(\theta t + \varphi_n\right), \qquad \varphi_n = \arg\left(a\,v_n\right)$$

- A damped wave with period $2\pi/\theta$, the same for every sector
- The phase φ_n depends on the sector: sectors oscillate at the same period but can peak in different months

$\Rightarrow$ Each conjugate pair is one real damped wave

A cycle of duration τ has τ eigenvalues

- On the cycle, after the shock: $\hat{y}_{n,t} = w\,\hat{y}_{n,t-\tau}$
- To know the future, we need the sector's output in each of the last τ months: those goods are still on their way back to it
- So the cycle has τ state variables, and τ eigenvalues

A cycle of duration τ has τ eigenvalues

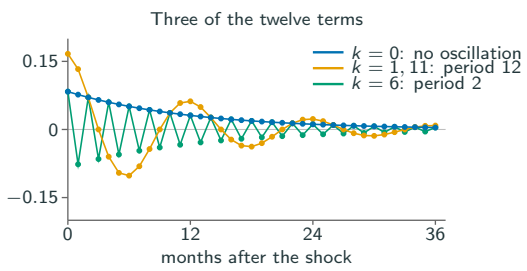
- On the cycle, after the shock: $\hat{y}_{n,t} = w\,\hat{y}_{n,t-\tau}$
- To know the future, we need the sector's output in each of the last τ months: those goods are still on their way back to it
- So the cycle has τ state variables, and τ eigenvalues
- Propulsion units: $\tau = 12$, twelve eigenvalues
- LP83: $\tau = 1$, one state variable, one eigenvalue

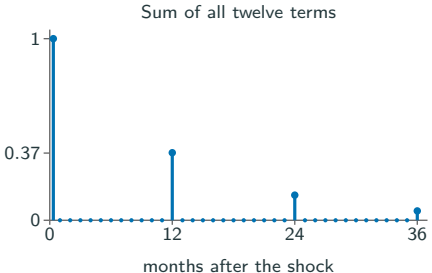
$\Rightarrow$ The number of roots counts the months the cycle has to remember

The twelve terms add up to one echo a year

- Shock at date 0: response 1, then w at month 12, w^2 at month 24, zero in between
- One term ξ^t never stays at zero for eleven months; the twelve terms together do:

$$\frac{1}{12}\sum_{k=0}^{11}\xi_k^t = w^{t/12}\cdot\frac{1}{12}\sum_{k=0}^{11}e^{2\pi ikt/12} = \begin{cases} w^{t/12} & \text{if 12 divides } t \\ 0 & \text{otherwise} \end{cases}$$





Each root is a harmonic of the yearly echo

- Root k has argument $2\pi k/\tau$: period τ/k months

The twelve roots of propulsion units

k	0	$1, 11$	$2, 10$	$3, 9$	$4, 8$	$5, 7$	6
Period (months)	none	12	6	4	3	2.4	2

- One positive real root, one negative real root, five conjugate pairs: seven periods
- $k = 1$ is the cycle's own period; the others are its harmonics
- They are the peaks of the cycle's spectrum, at τ/k

$\Rightarrow$ One cycle, one echo a year, seven periods in the spectrum

Long delays put the eigenvalues near the unit circle

- All τ roots have modulus $w^{1/\tau}$
- Per lap of the cycle, the response shrinks by w, whatever the delay
- Per month, it shrinks by $w^{1/\tau}$: the longer the delay, the closer to 1

Long delays put the eigenvalues near the unit circle

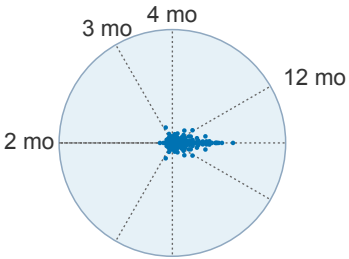
- All τ roots have modulus $w^{1/\tau}$
- Per lap of the cycle, the response shrinks by w, whatever the delay
- Per month, it shrinks by $w^{1/\tau}$: the longer the delay, the closer to 1
- Propulsion units, $w = 0.37$:
  - with $\tau = 1$ (LP83) the root would be 0.37: the response loses 63% a month
  - with $\tau = 12$ the modulus is 0.92: it loses 8% a month
- 0.92 is the largest modulus in the U.S. economy: these echoes die out last

$\Rightarrow$ Delays do not change the loss per lap; they stretch each lap over τ months

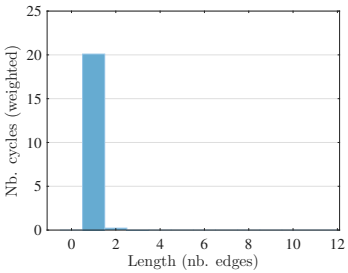
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From Network Cycles to Business Cycles (LP83)

Eigenvalues, LP83



Cycles by duration in LP83 (weighted)



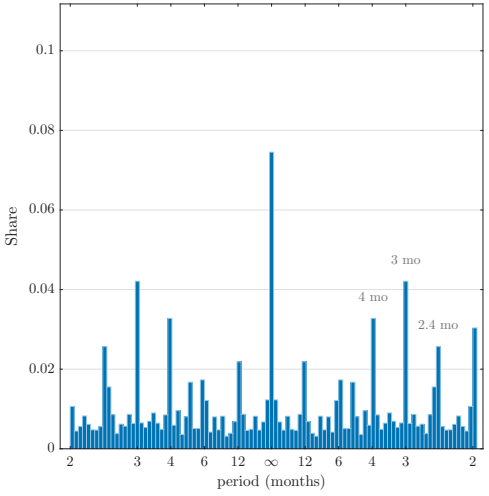
- Every delay is one month: a cycle's duration is its number of links
- Almost all the weight is at duration 1, the self-loops

- A one-month self-loop has one root, $\xi = \omega_{nn}$: real and positive, it does not oscillate
- The companion matrix is $\mathbf{\Omega}$ itself; its eigenvalues that are not positive real are small (Liu and Tsyvinski)

$\Rightarrow$ LP83 puts its cycles at a duration of one month, so it barely oscillates

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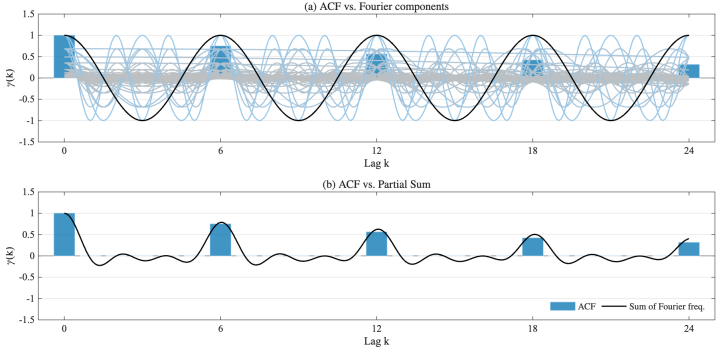
The periods of the eigenvalues



- Eigenvalues of the companion matrix, measured delays, by argument expressed as a period (months), weighted by modulus
- Center (∞): positive real eigenvalues, which do not oscillate
- Mass at 2, 3, 4 and 12 months, the durations of the heavy self-loops
- A cycle of duration τ also has roots at the periods τ/k: its harmonics

Back

Fundamentals and Harmonics



- A cycle with duration $\tau = 6$ and weight $w = 0.75$: autocorrelation w^j at lags $j\tau$, zero elsewhere (bars). (a) Its Fourier components, darker where the spectrum is larger; black: the fundamental, period τ. (b) Their sum rebuilds the autocorrelations
- The spectrum peaks at the fundamental and at its harmonics, periods τ/k

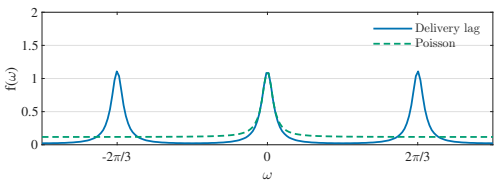
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Poisson Model

- A common way to model delays is Poisson arrival: for a delay τ, delivery with probability $1/\tau$ each period
- Example: sector n with a self-loop of weight w. Autocovariance of its output at lag $k \geq 1$:

$$\gamma_k = w \left(1 - \frac{1}{\tau}\right)^{k-1} \frac{1}{\tau} \sigma^2 (\hat{y}_{nt}) + \text{further laps}$$

With a delivery lag instead: $\gamma_k = w^j \sigma^2 (\hat{y}_{nt})$ at $k = j\tau$, zero at every other lag



Spectrum of a self-loop with $\tau = 3$ and $w = 0.75$: delivery lag vs. Poisson arrival

$\Rightarrow$ Poisson arrival spreads each lap over all later months: the peaks at $2\pi/\tau$ disappear

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Delay shocks: a disruption to deliveries

- Echoes do not need a productivity shock: a disruption to deliveries (2021-22) travels the same walks
- The static network has no goods in process and the LP83 pipeline is one month long: neither has room for such a shock
- *Delay shock of T months* to n: at date 0, a fraction ε of the goods due in the next T months will arrive T months late

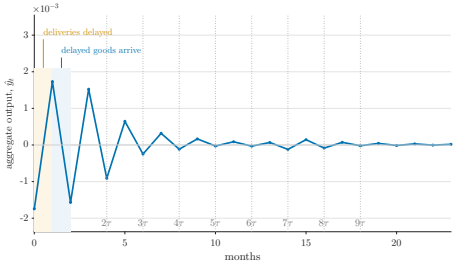
$$\hat{X}_{nj} = -\varepsilon \quad \text{for } j = 0, \ldots, T-1, \qquad \hat{X}_{nj} = +\varepsilon \quad \text{for } j = T, \ldots, 2T-1$$

  - X_{nj}: stock of good n that is j months from delivery
  - T is the length of the disruption; the delay τ_n itself is unchanged
- Nothing is created or destroyed: the goods leave the near end of the pipeline and reappear T months further out

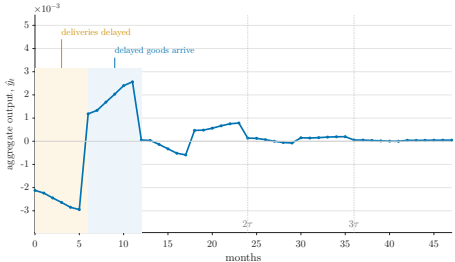
Delay shocks echo at the sector's delay

Response of aggregate output (value-added weighted), $\varepsilon = 1$.

Nonferrous metal, delay 2 months; $T = 1$



Motor vehicle parts, delay 12 months; $T = 6$



- Output falls while the deliveries are missing and rebounds when they arrive; the cumulative response is exactly zero
- The fall and rebound recur at the sector's delay: the echo of the first frame
- Trough: as deep as the impact of a same-size productivity fall (left), 1.6 times as deep (right)

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The spectrum of aggregate output: transfer form

Proposition

$$
\begin{aligned}
f_y(\omega) &= \tfrac{1}{2\pi}\boldsymbol{\mu}'\mathbf{M}(e^{-i\omega})\boldsymbol{\Sigma}\,\mathbf{M}(e^{-i\omega})^*\boldsymbol{\mu} \\
&= \sum_n \mu_n^2 f_n(\omega) + \tfrac{1}{2\pi}\sum_{n\neq m}\sum_k \mu_n\mu_m[\boldsymbol{\Gamma}_k]_{nm}e^{-i\omega k}
\end{aligned}
$$

- $\mathbf{M}(z) = (\mathbf{I} - \boldsymbol{\Omega}\,\mathbf{diag}(z^{\boldsymbol{\tau}}))^{-1}$: the sum of $w(\varsigma)z^{\tau(\varsigma)}$ over walks, so aggregate output is $\boldsymbol{\mu}'\mathbf{M}(L)\hat{\mathbf{A}}_t$; $\boldsymbol{\Sigma}$: covariance of the shocks; *: conjugate transpose
- First line: walks whose arrival dates align at a frequency reinforce each other there; the others offset each other
- Second line: the same quadratic form split into its diagonal terms (sectoral spectra) and its off-diagonal terms (comovements between sectors)

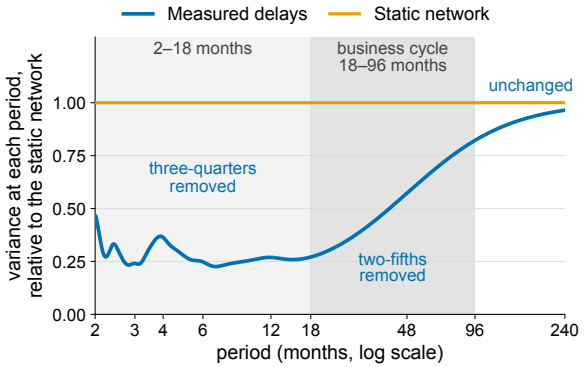
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An experiment: same network, same shocks, only the delays differ

- Four economies, same input-output matrix and long-run multiplier
  - no delays: the static network
  - one month everywhere: LP83
  - five months everywhere
  - the measured delays
- Same shocks in all four: persistent, as estimated (sectoral AR(1), common factor)
- Compute the spectrum of aggregate output in each
- Read it relative to the static network: 1 = the static network's variance at that period

$\Rightarrow$ How much of the network's business-cycle variance do the delays remove?

Delays damp aggregate output at business-cycle periods

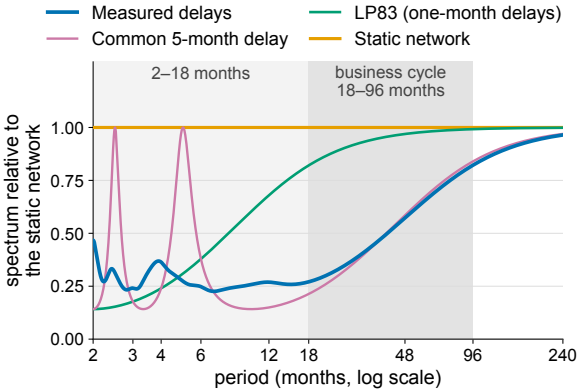


- Long periods: unchanged. Each shock has the same total effect, only spread over months
- Business cycle, 18–96 months: two-fifths of the variance removed
- 2–18 months: nearly three-quarters
- Why: arrivals spread over months partly cancel for movements faster than the spread

Overall variance: about 80% of the static network's. Most variance is in slow movements, which the delays leave alone.

$\Rightarrow$ A static reading overstates the network's business-cycle variance by 70%

Dispersed delays damp; one common delay resonates



- Common 5-month delay: every walk arrives at a multiple of 5 months. At the period of 5 months and its harmonic, 2.5, they line up: spikes above 1
- Measured delays are dispersed: arrivals never line up, only ripples
- LP83: one-month delays, too short to cancel at business-cycle periods

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With persistent productivity: our model near one, static at most zero

Why: the predictions 1 and 0 assume i.i.d. growth. Productivity is persistent in the data.

Setup: independent AR(1) productivities, persistence $\rho_n \in [0, 1)$; all delays ≥ 1 month.

Proposition

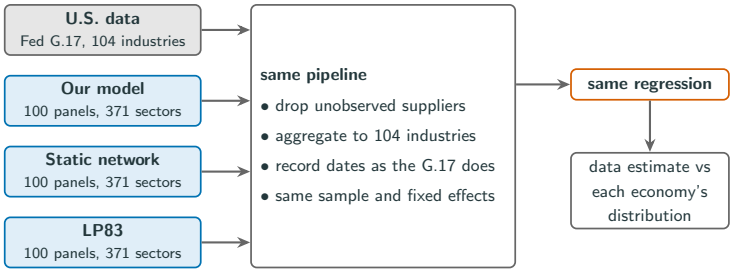
Our model: *the slope of* $\Delta \hat{y}_{nt}$ *on* $X_{nt}^{(\tau)}$ *is* $\beta_\tau = 1 + Cov(\Delta \hat{A}_{nt}, X_{nt}^{(\tau)}) / Var(X_{nt}^{(\tau)}) \leq 1$*; the gap is of order* $(1 - \rho_n)^2$ *times the weight of the loops through* n. **Static network**: *the slope is* ≤ 0*, on* $X^{(\tau)}$ *and on any composite with one common lag* $k \geq 1$.

- Why static ≤ 0: its growth is productivity growth; the composite is past growth; AR(1) growth has nonpositive autocovariance at every lag
- Holds with a common AR(1) factor whose loadings share a sign
- Why growth rates: in levels, persistence alone gives static slopes of 0.8 to 3
- TFP growth is near white noise: annual autocorrelation -0.03 (78 KLEMS series)

Back

Each economy is simulated and measured exactly like the data

- 100 simulated panels per economy, run through the same pipeline as the data: unobserved suppliers dropped, output recorded on the G.17's dates (dating-adjusted design)



- An economy's prediction: the distribution of the estimate across its 100 panels

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Two designs for the dates

- Naive: treat every series as built-dated. Data 0.33 (0.08); our model 0.99 as the work progresses, 0.14 at delivery
- Dating-adjusted: undo the worst case, a sale-dated index running a full delay behind production
- Sale-dated customer n: its index runs τ_n behind, so its response to a supplier shows up at $\tau_m + \tau_n$. **Shift it back** by τ_n
- Sale-dated supplier m: its index runs τ_m behind, so its customers respond to it at lag 0, where the static network also puts it. **Drop it**
- Same shifts and drops in the simulated economies
- It works under either timing: our model predicts 0.62 if sale-dated output is recorded as the work progresses, 0.65 at delivery
- So the panel test holds whatever the timing; the aggregate test is the one that picks it

Table of lags ◂ Back

How recording dates shift measured timing

- An index records output when it is built ($\delta = 0$) or at completion or sale ($\delta = \tau$). The measured series is production shifted by δ
- So customer *n* appears to respond to supplier *m* at $\tau_m + \delta_n - \delta_m$:

	Customer	
Supplier	built	sale
built	τ_m	$\tau_m + \tau_n$
sale	0	τ_n

- A sale-dated supplier shows at lag 0, where the static network also puts it: uninformative
- **Dating-adjusted design**: shift sale-dated customers by τ_n; weight each supplier by its built-dated share (46 indexes fully, 34 none, 24 partly)
- Same rules applied to the simulated economies

Fed's documentation: 57 built-dated, 31 at completion, 11 at shipment, 4 interpolated, 1 undocumented.

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Better-measured delays give a larger coefficient

- **What we do**: rerun the panel test with the Census-only delays: measured for the manufacturing core, one imputed value elsewhere

		Data		Simulated economies (mean)		
Design	Delays	$\widehat{\beta}_\tau$	(s.e.)	Our model	Static	LP83
Dating-adjusted	**Benchmark**	**0.91**	**(0.49)**	**0.65**	**−0.25**	**−0.12**
Naive	Benchmark	0.33	(0.08)	0.99	0.00	0.06
Dating-adjusted	Census only	0.77	(0.46)	0.68	−0.36	−0.12
Naive	Census only	0.27	(0.06)	1.00	0.00	−0.01

- Replacing imputed delays by measured ones: 0.77 to 0.91
- Why: a wrong lag misdates a supplier's whole contribution and attenuates the coefficient
- Every row: above all 100 static and all 100 LP83 draws

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The aggregate test under both recording readings

- All 104 industries stay: dropping the sale-dated ones would lose power and change the aggregate
- The simulations bracket what a sale date means: **production-timed** (books follow the work, the Census instruction); **worst case** (nothing recorded until delivery)

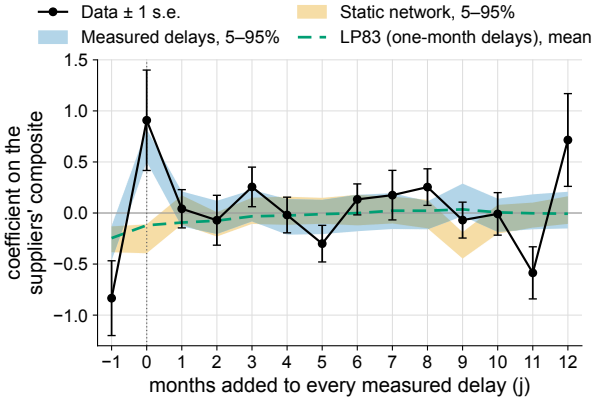
	$\widehat{\beta}_\tau$		Margin over best fixed lag	
Simulated economy	production-timed	worst case	production-timed	worst case
Our model	1.120 ± 0.124	0.116 ± 0.100	0.388 ± 0.116	-0.557 ± 0.112
Static	-0.124 ± 0.185	0.306 ± 0.133	-0.300 ± 0.189	0.005 ± 0.140
LP83	0.158 ± 0.141	0.155 ± 0.141	-1.099 ± 0.170	-1.124 ± 0.154
Data	**1.18**		+0.91 (excluding lag 12); +0.43 (including)	

- The data match the production-timed reading (1.12), not the worst case (0.12). Sample ending in 2019: 1.25, benchmarks unchanged
- Simulated margins exclude the lag-12 cell (no simulated economy has the data's lag-12 structure); the main slide quotes the data's +0.43, which includes it

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The comovement peaks at the measured delays

- **What we do**: add j months to every supplier's delay and rerun the dating-adjusted regression, $j = -2, \ldots, 12$. Right delays: a peak at $j = 0$



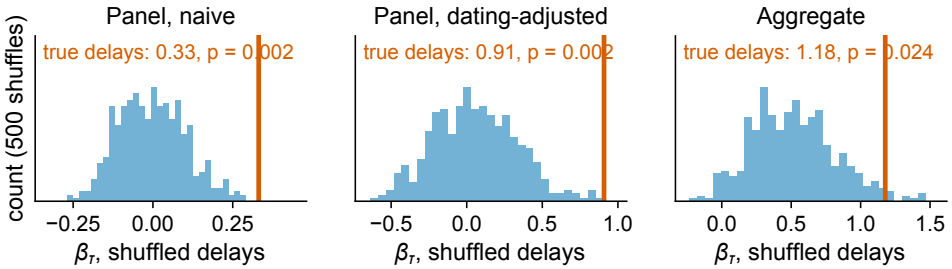
- Data: peak at $j = 0$, 2.4 data s.e. above the static band (4.4 in the naive design)
- Also high at $j = +12$, which no economy predicts
- The true delays beat all 500 random reassignments; imputed delays carry nothing; robust to noise

Permutations Noise

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Shuffled delays carry no signal

- **What we do**: reassign the measured delays at random across the 371 sectors, 500 times; rerun the three tests each time; compare with the true delays



- Panel tests: shuffled coefficients center at zero; the true delays beat all 500 ($p = 0.002$)
- Aggregate: shuffled coefficients center at 0.52. Without time effects any dispersed delay vector inherits aggregate momentum. The true delays beat 489 of 500
- So the aggregate identification rests on the margin over fixed lags

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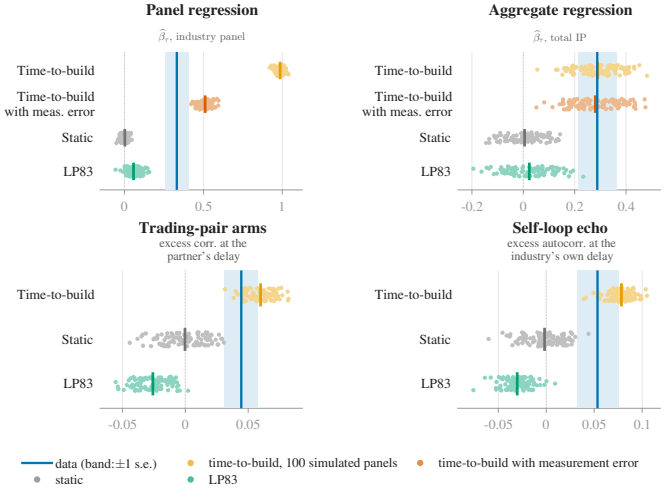
Measurement noise does not rescue the static network or LP83

- **What we do**: add classical measurement error to every simulated series, rerun the tests
- Calibration: simulated growth variance matches the data's, an extreme reading (about 80% of industry residual variance called noise); scaled by 0.5, 1 and 2
- In every design and at every scale: the data estimate exceeds all 100 static and all 100 LP83 draws
- Noise also lowers our model's prediction. Noise this large would attenuate the data's estimate to 0.1–0.2, below the 0.33–0.91 observed

$\Rightarrow$ The residual variance is better read as non-productivity variation in output

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Each test favors our model over the static network and LP83



- Blue line: the estimate in the data, band ±1 standard error
- Dots: the same estimate on 100 simulated panels per economy; "Time-to-build" is our model
- No parameter is fitted to these data

⇒ Shocks travel through the U.S. network with the measured delays

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